HealthConnect Rwanda

Healthcare Operations & Performance Analytics Report

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Executive Summary: HealthConnect Rwanda is a multi-facility healthcare provider committed to delivering accessible, high-quality medical services across its network. To support evidence-based strategic decision-making, executive leadership commissioned an end-to-end data analytics workflow spanning database engineering in PostgreSQL, exploratory data analysis, and interactive business intelligence dashboard development in Power BI. This report synthesizes operational, financial, and clinical insights across **20 healthcare facilities**, **350 active physicians**, **5,000 recorded patient appointments**, and **4,400 registered patients**. Overall, the organization generated **$9.3 Billion in Revenue** with an average patient length of stay of **5.49 days**.

1. Executive Key Performance Indicators (KPIs)

The executive management dashboard tracks six core organization-wide metrics to evaluate system capacity, medical workforce, patient volume, and overall financial performance:

EXECUTIVE KPI	VALUE / METRIC	OPERATIONAL CONTEXT & SIGNIFICANCE
Total Revenue	**$9.3 Billion**	Cumulative revenue generated across all facilities and diagnostic tests.
Total Registered Patients	**4,400 Patients**	Unique patient database volume across all regional coverage areas.
Total Appointments	**5,000 Visits**	Total outpatient consultations, inpatient admissions, and follow-up visits.
Total Doctors	**350 Physicians**	Active medical staff roster distributed across clinical departments.
Total Healthcare Facilities	**20 Facilities**	Total operational hospital sites and clinics across provinces.
Average Length of Stay	**5.49 Days**	Mean turnaround duration for inpatient stay and bed utilization.

2. Departmental & Facility Workload Analysis

Facility Volume & Revenue Performance

Patient volume and financial performance show high concentration across top-tier facilities. **Facility HF001** and **Facility HF002** emerge as the highest revenue generators, driven by high outpatient appointment throughput and diagnostic lab test volume.

Departmental Distribution & Bottlenecks

- **Top Busiest Departments:** Cardiology, Radiology, Obstetrics & Gynecology, and Laboratory consistently record the highest appointment volume.
- **Workload Imbalance:** Cardiology and Radiology handle over 30% of total patient visits, creating peak-hour scheduling bottlenecks and extended waiting times.

- **Medical Staff Distribution:** While physician distribution across Radiology, Cardiology, and ICU aligns relatively well with visit volume, secondary facilities experience occasional doctor shortages during high-demand months.

3. Clinical & Diagnostic Insights

Disease Prevalence & Diagnosis Breakdown

Analysis of diagnostic records across registered patients reveals five predominant conditions:

1. **Hypertension** *(Leading chronic diagnosis)*
2. **Gastritis**
3. **Upper Respiratory Infections (URI)**
4. **Diabetes**
5. **Malaria**

Hypertension and Diabetes represent the vast majority of chronic follow-up appointments, indicating a critical need for preventative community healthcare initiatives and specialized chronic care management programs.

Clinical Laboratory & Diagnostic Performance

- **Total Lab Test Types:** 4 standardized diagnostic categories.
- **Lab Outcome Status:** Laboratory test results are categorized into **Normal** and **Abnormal** outcomes, with abnormal findings triggering mandatory specialist consultations in Cardiology and Internal Medicine.
- **Inpatient Turnaround Time:** The average length of stay across rooms stands at **5.49 days**, with ICU and Emergency rooms recording higher stay durations compared to routine observation wards.

4. Financial Performance & Revenue Drivers

Revenue Contribution by Specialty & Department

- **Cardiology** and **Surgery** generate the largest proportion of total institutional revenue, backed by high-value diagnostic procedures and surgical interventions.
- **Laboratory & Diagnostic Services** act as high-volume, steady revenue contributors, processing 5,000 individual lab consultations.
- **Revenue Trend:** Revenue fluctuates seasonally, peaking in **March** before stabilizing across mid-year quarters.

Insurance Coverage & Cost Structure

The majority of patient appointments are supported by national and private health insurance schemes. Maintaining strong relationships with major insurance providers is critical for sustaining prompt claim reimbursements and minimizing outstanding receivables.

5. Strategic Recommendations

1. **Facility Capacity Expansion:** Allocate capital expenditure toward expanding inpatient bed capacity and consultation rooms in **Facility HF001** and **HF002** to alleviate bed occupancy constraints.
2. **Clinical Resource Reallocation:** Deploy additional medical officers and specialist nurses to **Cardiology** and **Radiology** during peak operational months (January and March) to reduce patient waiting times.

3. **Chronic Disease Management Programs:** Establish specialized outpatient clinics for **Hypertension** and **Diabetes** to improve long-term patient outcomes and reduce acute emergency readmissions.
4. **Revenue Cycle Optimization:** Streamline billing workflows for high-revenue service categories (Surgical and Diagnostic procedures) to reduce reimbursement claim cycles from insurance providers.
5. **Length-of-Stay Optimization:** Standardize post-treatment discharge protocols in general observation rooms to reduce the average length of stay from 5.49 days to under 4.5 days, freeing up room availability for critical admissions.